Network Modeling Homework:
Simulation of Traffic Sources and Composite Traffic Shaper

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Assignment

The objective of this homework is to implement a composite traffic shaper consisting of a Token Bucket Filter (TBF) followed by a Leaky Bucket Filter (LBF), and to develop a simple simulation to investigate its performance when processing the traffic generated by the source model developed in Homework 1.

You are required to develop two software modules:

- *Source module*: An application that generates a synthetic packet flow according to the traffic model developed in Homework 1, reproducing the distributions of packet generation times, batch sizes, and packet lengths. The generated traffic shall be saved in a file, named Input_Traffic_Trace.txt, with the format described later.
- *Shaper module*: Another software module that reads the traffic trace file generated by the Source Model and processes it through a composite shaper consisting of a TBF followed by an LBF. The resulting output traffic shall be saved in a file named Output_Traffic_Trace.txt having the same format of the other trace file.

The TBF is characterized by the following parameters:

- Token generation rate ρ (tokens/s).
- Token bucket capacity b (tokens).
- Input buffer capacity B_i (bytes).

Assume that one token permits the forwarding of one byte of data. A packet can leave the TBF only when the bucket contains enough tokens to cover its entire length. When the packet is forwarded, the corresponding number of tokens is removed from the bucket. The token bucket capacity shall be sufficiently large to accommodate the largest packet generated by the source. Alternatively, the implementation shall explicitly define how packets larger than the bucket capacity are handled.
Packets that cannot be forwarded immediately because of insufficient tokens shall be queued in the input buffer and served according to a First-In First-Out (FIFO) policy.
The LBF is characterized by the following parameters:

- Maximum forwarding rate μ (bytes/s).
- Output buffer capacity B_o (bytes).

Packets arriving at the LBF are buffered in the output buffer and forwarded at rate μ, according to a FIFO policy. The transmission of a packet of length L bytes requires L/μ seconds. Packet transmissions are non-preemptive.
Packets that cannot be accommodated in either buffer shall be discarded.
All five parameters of the composite shaper shall be configurable to investigate their effects on the output traffic and application QoS.

Implementation details

Each generated packet shall be assigned a unique identifier, which shall be preserved throughout the system.
To simplify the implementation, packets may be represented by descriptors containing their identifier, generation timestamp, and length, without transmitting the actual payload.
The traffic shaping process shall operate in simulated time. In particular, packet generation, token accumulation, buffering, and forwarding shall be governed by a logical simulation clock, rather than by the actual execution time of the software.

Traffic traces

The input and output traffic traces shall be saved in two separate files. Each file shall contain one row per packet, reporting:

- Packet identifier.
- Timestamp (relative to the beginning of the experiment).
- Packet length (bytes).

The timestamp shall represent the packet generation time in the input trace and the completion time of packet forwarding in the output trace.
Packets generated in the same batch shall appear on separate rows of the input trace, with identical generation timestamps.
Packets discarded by the shaper shall not appear in the output trace.
The packet identifiers shall allow the input and output traces to be matched in order to determine packet delays and losses.
Additionally, separate trace files can be maintained to trace buffering at the input and output buffer, if deemed useful for the analysis.

1 Experiments

1. *QoS indicators*: Identify a set of Key Performance Indicators (KPIs) suitable for assessing the Quality of Service (QoS) of the application modeled in Homework 1.

   Possible indicators include throughput, mean packet delay, delay percentiles, jitter, and packet loss rate.

   Justify your selection based on the characteristics and requirements of the target application you considered in Homework 1.

2. *QoS requirements*: For each selected KPI, define a target value or acceptable range corresponding to satisfactory QoS for the considered application. Briefly justify the selected thresholds.

3. *Statistical analysis*: Run the experiments for a sufficiently long period to obtain statistically meaningful estimates of the selected KPIs.

   Justify the duration of the experiments and discuss the statistical reliability of the results.

4. *Sensitivity analysis*: Explore the combinations of shaper parameters that satisfy the QoS requirements defined in Step 2.

   Identify the trade-offs between buffer capacities, token generation rate, token bucket capacity, and maximum forwarding rate, and determine a resource-efficient configuration.

   Explain and justify the criteria adopted to select the configuration.

   Present and discuss the results using appropriate plots and statistical measures.

5. *Report*: Write a report of **max 3 pages** describing the implementation, experimental methodology, results, and conclusions.

   Discuss the main trade-offs between traffic shaping, resource requirements, and application QoS.

2 Deliverables

Submit the following:

- The source code of the Source and Shaper modules, including instructions for compiling and running the applications.
- One Input_Trace_File.txt and its corresponding Output_Trace_File.txt
- The report in PDF format describing the implementation, experiments, results, and conclusions.